

Group Study Room with Chat

Final Year Project Proposal

Session 2025-2027

A project submitted in partial fulfillment of
The Islamia University of Bahawal Pur

Of
BS In Information Technology



Hoorain Hasnain

S25BINFT7M04009

Supervisor Sir Faisal Shahzad

Department of Information Technology

Faculty of Computing

Islamia University of Bahawal Pur

Project Title: Group study Room with Chat

Introduction:

The **Group Study Room with Chat** is a simple web-based application designed for students to collaborate in virtual study rooms. It allows students to create or join a study room, communicate with peers, and share study materials via a built-in chat feature. This platform replicates the traditional group study experience, where students can ask questions, exchange ideas, and study together in a digital space.

Objective:

The main goal of this project is to:

- Provide a **virtual space** for group study sessions.
- Enable **real-time communication** through a chat feature.
- Allow users to **share study materials** (documents, notes, links).
- Create a user-friendly platform that is easy to use and navigate.

Scope:

1. **Room Creation and Joining:**
 - a. Users can create a study room and invite others to join via a unique link.
 - b. Participants can join public or private study rooms.
2. **Real-Time Chat:**
 - a. Users can chat with others in the room in real-time.
 - b. Supports **text, images, and file sharing**.
3. **File Sharing:**
 - a. Users can upload and share study materials like PDFs, documents, and links within the room.
4. **User Authentication:**
 - a. Simple sign-up and login system to ensure only registered users can access the study rooms.

Technology Stack:

1. **Frontend:**
 - a. **HTML** for page structure.
 - b. **CSS** for styling.
 - c. **JavaScript** (Vanilla JS or React) for interactivity.
2. **Backend:**
 - a. **Node.js** with **Express** for handling server-side logic.
 - b. **Socket.io** for enabling real-time chat functionality.
3. **Database:**
 - a. **MongoDB** for storing user data, study rooms, and chat history.
4. **Authentication:**
 - a. Simple **JWT** (JSON Web Tokens) for user authentication.

Expected Outcome:

1. **Easy Communication:** Students can easily communicate with peers in study groups, share materials, and discuss topics.
2. **Improved Collaboration:** Group study becomes more effective through real-time chat and resource sharing.
3. **Simple User Interface:** A minimalistic, user-friendly interface will make it easy for anyone to use without technical knowledge.
4. **Growth Opportunity:** The platform can be expanded in the future to include features like voice chat, video calls, or virtual whiteboards.